

# Pénélope Azuelos

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## Education

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| 2022 – present | University of Bristol, UK<br><i>PhD in Mathematics (ongoing), supervised by Mark Hagen and Indira Chatterji</i> |
| 2021 – 2022    | École Normale Supérieure de Lyon, France<br><i>Master (M2) Mathématiques Avancées</i>                           |
| 2017 – 2021    | University of Bristol, UK<br><i>Master in Science in Mathematics (with first class honours)</i>                 |

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## Publications

(with Damien Gaboriau) *Perfect kernel and dynamics: from Bass-Serre theory to hyperbolic groups*, *Mathematische Annalen* 391: 4733–4789, 2025.

<https://doi.org/10.1007/s00208-024-03038-w>

*On subgroups with narrow Schreier graphs*, *Bulletin of the London Mathematical Society*, 56(12): 3652–3668, 2024. <https://doi.org/10.1112/blms.13157>

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## Preprints

(with Mark Hagen) *Periodic quasiflats in hierarchically hyperbolic spaces*, arXiv:2608.01513.

*Uncountably many homogeneous real trees with the same valence*, arXiv:2511.03722.

*A guide to constructing free transitive actions on median spaces*, to appear in *Advances in Mathematics*, arXiv:2507.22230.

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## Teaching

*Teaching assistant at the University of Bristol for the following courses:*

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|-------------|---------------------------------------------------|
| Spring 2024 | Introduction to proofs and group theory MATH10010 |
| Autumn 2023 | Combinatorics MATH30030                           |
| Spring 2020 | Introduction to geometry MATH20004                |

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## Conference Talks

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| June 2026  | Groups acting on non-positively curved spaces, International Academic Forum, Heidelberg, Germany |
| –          | Branch groups: subgroups, rigidity, topologies, Universidad Complutense de Madrid, Spain         |
| May 2026   | Geometric group theory around cube complexes, CIRM, France                                       |
| April 2026 | Coarse geometry of groups, Centre Paul-Langevin, Aussois, France                                 |

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|----------------|---------------------------------------------------------------------------------------------------------------|
| February 2026  | Median Geometry and Applications, MFO, Oberwolfach, Germany.                                                  |
| –              | Orbit meetings, University of Warwick, UK                                                                     |
| November 2025  | Parole aux jeunes scientifiques du RT tétraèdre de Platon, Montpellier, France.                               |
| –              | Operator algebras in the south of England, University of Bristol, UK.                                         |
| June 2025      | Conference on hyperbolicity and generalizations, Harbin Institute of Technology, China.                       |
| January 2025   | RSME's 7th congress of young researchers, parallel session on geometric group theory, UPV/EHU, Bilbao, Spain. |
| September 2024 | World of Geometric Group Theory, online.                                                                      |
| March 2024     | Rigidité en théorie des groupes : aspects géométriques et ergodiques, Université Paris-Saclay, Orsay, France. |
| May 2023       | Measured Group Theory, Stochastic Processes on Groups and Borel Combinatorics, CIRM, Marseille, France.       |

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### Seminar Talks

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| May 2026       | Geometric group theory seminar, University of Oxford, UK                                           |
| April 2026     | Junior geometry seminar, University of Bristol, UK                                                 |
| March 2026     | Junior topology and group theory seminar, University of Oxford, UK                                 |
| February 2026  | DAGGER seminar, University of Warwick, UK.                                                         |
| –              | Junior Geometry Colloquium, KIT, Karlsruhe, Germany.                                               |
| January 2026   | Geometry and Topology Seminar, University of Bristol, UK.                                          |
| October 2025   | Junior Algebra and Combinatorics Seminar, University of Bristol, UK.                               |
| September 2025 | Séminaire Groupes et Actions, Université Paris-Saclay, Orsay, France.                              |
| February 2025  | Junior Geometry Seminar, University of Bristol, UK.                                                |
| December 2024  | Séminaire d'Algèbres d'Opérateurs, IMJ-PRG, Paris, France.                                         |
| October 2024   | Séminaire de l'équipe d'algèbre, topologie et géométrie, Laboratoire J.A. Dieudonné, Nice, France. |
| September 2024 | Séminaire de Théorie Ergodique, Institut de recherche mathématique de Rennes, France.              |
| –              | Algebra Geometry and Topology seminar, Heriot-Watt University, Edinburgh, UK.                      |
| May 2024       | International Young Seminar on Bounded Cohomology and Simplicial Volume, online.                   |
| March 2024     | GdT Action ! ENS de Lyon, France.                                                                  |
| February 2024  | Junior Geometry Seminar, University of Cambridge, UK.                                              |
| December 2023  | Postgraduate Seminar, University of Bristol, UK.                                                   |
| November 2023  | Junior Topology and Group Theory Seminar, University of Oxford, UK.                                |
| September 2023 | Le séminaire virtuel francophone Groupes et Géométrie, online.                                     |
| January 2023   | Junior Geometry Seminar, University of Bristol, UK.                                                |

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## Service

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| 2025        | Organisation of the Postgraduate Group Theory Conference 2025 |
| 2023 – 2024 | Organisation of the Bristol Junior Geometry Seminar           |

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## Outreach

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| July 2026 | Talk and panel participation at an undergraduate summer school for women in mathematics, University of Warwick |
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## Awards and funding

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| 2024        | Grant from the Académie d'Excellence "Systèmes Complexes" for a three month research visit to the LJAD in Nice, supervised by Indira Chatterji. |
| 2022 – 2026 | University of Bristol PhD scholarship.                                                                                                          |
| 2021 – 2022 | MiLyon scholarship for an M2 masters program.                                                                                                   |
| 2020        | Summer research bursary funded by the Heilbronn Institute for Mathematical Research, supervised by Misha Rudnev.                                |

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## Languages

English and French (native speaker)